

Kenya Certificate of Secondary Education (K.C.S.E).

121/2

MATHEMATICS

PAPER 2

TIME: 2 ½ HOURS

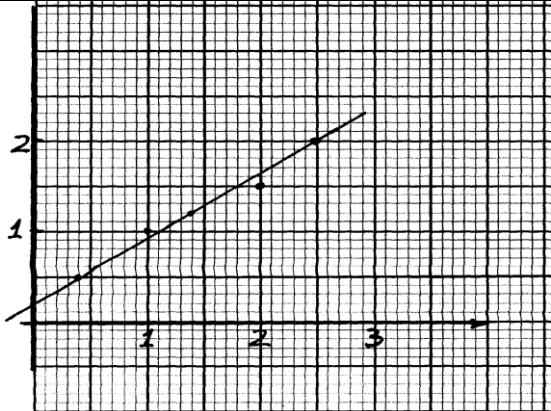
MARKING SCHEME

NO.	WORKING	MARKS	REMARKS
1.	$= \log 5^2 - \log 16^{\frac{1}{2}} + \log 40^2$ $= \log 25 - \log 4 + \log 1600$ $= \log \left(\frac{25 \times 1600}{4} \right)$ $= \log 10,000$ $= 4$	M1 M1 A1	Apply law $n \log A = \log A^n$ Combining logs For ✓ answer
		3mks	
2.	$2x - (x - 3) = 0$ $2x - x + 3 = 0$ $x = -3$	M1 A1	For equating determinant to zero For ✓ value of x
		2mks	
3.	$a^4 = \frac{1+d^2}{b^2} + \frac{b}{3}$ $a^4 - \frac{b}{3} = \frac{1+d^2}{b^2}$ $b^2 \left(a^4 - \frac{b}{3} \right) = 1 + d^2$ $b^2 \left(a^4 - \frac{b}{3} \right) - 1 = d^2$ $d = \sqrt{b^2 \left(a^4 - \frac{b}{3} \right) - 1}$	M1 M1 A1	For squaring both sides For d^2 For d
		3mks	
4.	$= \frac{3\sqrt{7} + \sqrt{7} - 2}{\sqrt{7}(\sqrt{7} - 2)}$ $= \frac{4\sqrt{7} - 2}{7 - 2\sqrt{7}}$ $= \frac{4\sqrt{7} - 2}{7 - 2\sqrt{7}} \times \frac{7 + 2\sqrt{7}}{7 + 2\sqrt{7}}$ $= \frac{4\sqrt{7} + (7 + 2\sqrt{7}) - 2(7 + 2\sqrt{7})}{21}$	M1 M1	For conjugate For rationalizing denominator



	$= \frac{28\sqrt{7} + 56 - 14 - 4\sqrt{7}}{21}$ $= \frac{42 + 24\sqrt{7}}{21}$ $= 2 + \frac{8}{7}\sqrt{7}$	A1	
		3mks	
5.	$height = \sqrt{15^2 - 9^2} = 12.0cm$ $v = \frac{1}{3} \times \pi \times 9.0 \times 9.0 \times 12.0$ $\% error = \left(\frac{0.05}{9.0} + \frac{0.05}{9.0} + \frac{0.05}{12.0} \right) \times 100$ $= 1.528\%$	B1 M1 A1	For height
		3mks	
6.	$A = k + \frac{m}{B}$ $-10 = k + \frac{m}{2.5} \quad (i)$ $-25 + 2.5k + m$ $10 = k + \frac{m}{1.25} \quad (ii)$ $12.5 = 1.25k + m$ $-25 = 2.5k + m$ $\underline{12.5 = 1.25k + m}$ $-37.5 = 1.25k$ $k = -30$ $12.5 = 1.25(-30) + m$ $m = 50$ $A = -30 + \frac{50}{B}$ $= -30 + \frac{50}{1.5}$ $= 3\frac{1}{3}$	M1 M1 M1 A1	For simultaneous equations for ✓ attempt to solve simultaneous equations.
		4 mks	
7.	$Area = h (y_1 + y_2 + \dots)$ $= 0.4 (6.2 + 4.3 + 2.6)$ $= 5.24$	M1 A1	for substitution for ✓ answer (CAO)
		2mks	
8.	$= 14 \times \frac{18}{10} \times \frac{30}{27}$ $= 28 \text{ pple}$	M1 M1 A1	for $\times \frac{18}{10}$ for $\times \frac{20}{27}$ for 28 pple
		3mks	



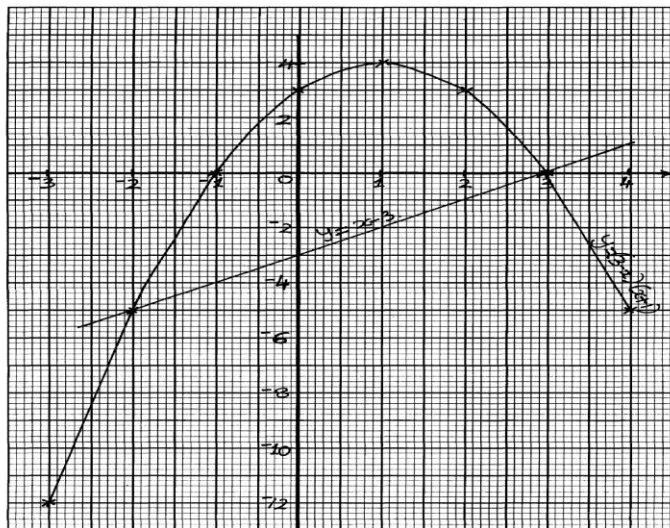
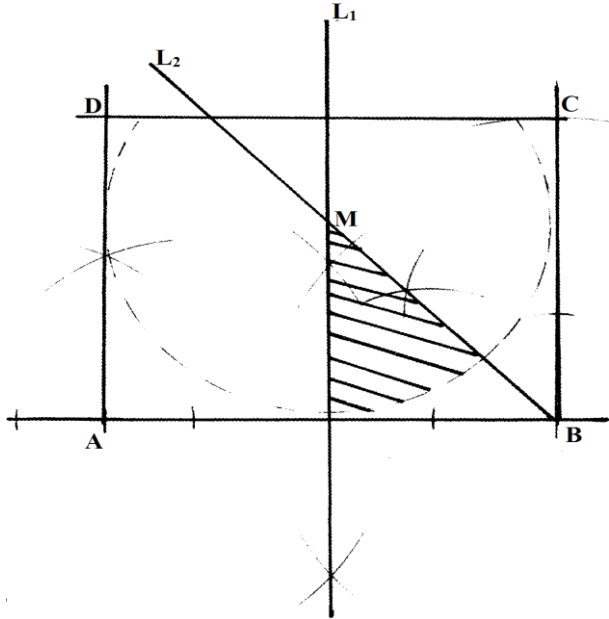
9.	(i) $35+15 = 50^0$ (ii) $Time = \frac{60 \times 50 \times \cos 60}{250}$ $= 6\text{hrs}$	B1 M1 A1	
		3mks	
10.	(a) $= x^5 + 5x^4(-0.2) + 10x^3(-0.2)^2 + 10x^2(-0.2)^3 + 5x(-0.2)^4$ $= x^5 - x^4 + 0.4x^3 - 0.08x^2 + 0.008x - 0.00032$ (b) $= 10^5 - 10^4 + 0.4(10)^3 - 0.08(10)^2$ 90392	M1 A1 M1 A1	
		4mks	
11.	(a) AC (b) $AC = \sqrt{12^2 + 5^2} = 13\text{cm}$ $\tan \theta = \frac{6.5}{13}$ $\theta = 26.57^0$	B1 B1 M1 A1	At 13 for $\tan \theta$ for θ
		3mks	
12.	$= \frac{\sin x \cos x + \sin x}{\cos x}$ $= \frac{\sin x \cos x}{\cos x} + \frac{\sin x}{\cos x}$ $= \sin x + \tan x$	M1 M1 A1	
		3mks	
13.	$A(-1,1,0)$ $B(x,y,z)$ $\Rightarrow \frac{-1+x}{2} = 1, \frac{1+y}{2} = -1.5, \frac{0+z}{2} = 2$ $x = 3, y = -4, z = 4$ $B(3,-4,4)$ $/OB/ = \sqrt{3^2 + (-4)^2 + 4^2}$ $= \sqrt{41}$ $= 6.403 \text{ units}$	M1 A1 M1 A1	Can be implied (accept column form)
		4mks	
14.	 $y = 0.7x + 0.2$	P1 L1 B1	
		3mks	



15.	2600 40 70 65 55 10 15 2:3	B1 B1 A1	for equivalent																																																								
		3mks																																																									
16.	$x(x+9)=36$ $x^2+9x-36=0$ $(x+12)(x-3)=0$ $x=-12, \quad x=3$ $BC=3cm$	M1 M1 B1																																																									
		3mks																																																									
17.	(a) (i) $250,000\left(1+\frac{4}{100}\right)^{10}$ $=sh.370061.0712$ (ii) $\frac{370,000}{4}$ $=sh. 92,500$ (iii) $10\,000\times 4\times 36$ $=sh. 1\,440\,000$ (b) (i) $92500\left(1-\frac{20}{100}\right)^3$ $=sh. 47\,360$ (ii) Profit = $(61+1440\,000+189\,440)-250\,000$ $=1379\,501$	M1 A1 M1 A1 M1 A1 M1 A1 M1 A1	Accept 370061.10																																																								
		10mks																																																									
18.	<table><tr><th>Class</th><th>x</th><th>tx-45.5</th><th>f</th><th>ft</th><th>ft²</th><th>cf</th></tr><tr><td>21-30</td><td>25.5</td><td>-20</td><td>15</td><td>-300</td><td>6000</td><td>15</td></tr><tr><td>31-40</td><td>35.5</td><td>-10</td><td>11</td><td>-110</td><td>1100</td><td>26</td></tr><tr><td>41-50</td><td>45.5</td><td>0</td><td>17</td><td>0</td><td>0</td><td>43</td></tr><tr><td>51-60</td><td>55.5</td><td>10</td><td>4</td><td>40</td><td>400</td><td>47</td></tr><tr><td>61-70</td><td>65.5</td><td>20</td><td>2</td><td>40</td><td>800</td><td>49</td></tr><tr><td>71-80</td><td>75.5</td><td>30</td><td>1</td><td>30</td><td>900</td><td>50</td></tr><tr><td></td><td></td><td></td><td>50</td><td>-300</td><td>9200</td><td></td></tr></table> (a) 31-40 (b) (i) $Mean=\frac{-300}{50}+45.5$ $=39.5yrs$ $s.d=\sqrt{\frac{9200}{50}-\left(\frac{-300}{50}\right)^2}$ $=12.17$ (c) $6^{th}=40.5+\left(\frac{30-26}{17}\right)10$ $=42.85$	Class	x	tx-45.5	f	ft	ft ²	cf	21-30	25.5	-20	15	-300	6000	15	31-40	35.5	-10	11	-110	1100	26	41-50	45.5	0	17	0	0	43	51-60	55.5	10	4	40	400	47	61-70	65.5	20	2	40	800	49	71-80	75.5	30	1	30	900	50				50	-300	9200		 B1 B1 M1 A1 B1 M1 A1 B1 M1 A1	For ✓median class For ✓ ft column For ✓ft ² column For ✓ c.f column
Class	x	tx-45.5	f	ft	ft ²	cf																																																					
21-30	25.5	-20	15	-300	6000	15																																																					
31-40	35.5	-10	11	-110	1100	26																																																					
41-50	45.5	0	17	0	0	43																																																					
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71-80	75.5	30	1	30	900	50																																																					
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19.	(a) $a+2d, \quad a+8d, \quad a+24d$ (b) (ii) $\frac{a+8d}{a+2d} = \frac{a+24d}{a+8d} \quad I$ $a+6d+2(a+5d)=78 \quad II$ $\left. \begin{array}{l} 16d=10a \quad I \\ 3a+16d=78 \quad II \end{array} \right\}$ $3a+10a=78$ $13a=78$ $a=6$ $16d=10 \times 6$ $d=3.75$ (ii) $S_9 = \frac{9}{2}(2 \times 6 + (9-1)3.75)$ $=189$ (ii) $\Rightarrow [6+6(3.75)] - [6+3(3.75)]$ $=11.25$	B1 M1 M1 M1 M1 A1 A1 M1 A1	For substitution or equivalent
		10mks	
20.	$P(A) = \frac{3}{4} \quad P(A^1) = \frac{1}{4}$ $P(B) = \frac{2}{3} \quad P(B^1) = \frac{1}{3}$ $P(C) = \frac{4}{5} \quad P(C^1) = \frac{1}{5}$ (a) $P(ABC) = \frac{3}{4} \times \frac{2}{3} \times \frac{4}{5}$ $= \frac{2}{5}$ (b) $P(A^1B^1C^1) = \frac{1}{4} \times \frac{1}{3} \times \frac{1}{5}$ $= \frac{1}{60}$ (c) $\Rightarrow P(A^1B^1C^1 \text{ or } A^1BC^1 \text{ or } A^1B^1C)$ $= \left(\frac{3}{4} \times \frac{1}{3} \times \frac{1}{5}\right) + \left(\frac{1}{4} \times \frac{2}{3} \times \frac{1}{5}\right) + \left(\frac{1}{4} \times \frac{2}{3} \times \frac{4}{5}\right)$ $= \frac{9}{60}$ (d) $\Rightarrow P(ABC^1 \text{ or } AB^1C \text{ or } A^1BC)$ $= \left(\frac{3}{4} \times \frac{2}{3} \times \frac{1}{5}\right) + \left(\frac{3}{4} \times \frac{1}{3} \times \frac{1}{5}\right) + \left(\frac{1}{4} \times \frac{2}{3} \times \frac{4}{5}\right)$ $= \frac{26}{60}$ (e) $\Rightarrow 1 - P(ABC)$ $= 1 - \frac{2}{5}$ $= \frac{3}{5}$	M1 A1 M1 A1 M1 A1 M1 A1 M1 A1	
		10mks	

21.	(a) <table border="1"><tr><td>x</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>x+1</td><td></td><td></td><td>0</td><td></td><td>2</td><td></td><td>5</td></tr><tr><td>3-x</td><td></td><td></td><td></td><td>3</td><td></td><td></td><td>0</td></tr><tr><td>y</td><td></td><td></td><td>0</td><td></td><td></td><td>3</td><td></td></tr></table> (b)  (c) (i) $(3-x)(x+1) = 0$ $-x^2 + 2x + 3 = 0$ $x = 3, -1$ (ii) $-x^2 + 2x + 3 = y$ $-x^2 + x + 6 = 0$ $x - 3 = y$ $x = -2, 3$	x								x+1			0		2		5	3-x				3			0	y			0			3		B₂ S₁ P₁ C B1B1 M1 L1 B1	For all ✓ values Allow B₁ for any four ✓ values ✓ scale ✓ plotting ✓ curve For y = x-3 For ✓ line drawn For both values
x																																			
x+1			0		2		5																												
3-x				3			0																												
y			0			3																													
22.	 $\frac{44}{360} \times \frac{22}{7} \times 3.5 \times 3.5$ $= 4.7 \text{ cm}_2$	10mks B1 B1 B1 B1 B1 B1 B1 B1 M1 A1	Angle bisector drawn ✓ly locating pts A,B,C,D ✓rectangle drawn L1 constructed L2 constructed Angle measured ✓ly $44 \pm 1^\circ$ An arc broken drawn inside rectangle radius 3.5cm ✓ region shaded																																
		10mks																																	



23.	(a) $x^2 + 5 = 8 - 2x$ $x^2 + 2x - 3 = 0$ $(x + 3)(x - 1) = 0$ $x = -3$ or 1 When $x = -3$, $y = 14$ When $x = 1$ $y = 6$ coordinates $C(-3, 14)$ and $D(1, 6)$ (b) $\int_{-3}^1 (x^2 + 5) dx$ $= \left[\frac{x^3}{3} + 5x + c \right]_{-3}^1$ $= \left(\frac{1}{3} + 5 \right) - \left(\frac{-3^3}{3} + 5(-3) \right)$ $= 32 \frac{1}{3}$ square units (c) Area under line $y = 8 - 2x$ $= \left(8x - \frac{2x^2}{2} \right)_{-3}^1$ $= (8(1) - 1^2) - (8(-3) - (-3)^2)$ $= 40$ (d) Shaded area $= 40 - 32 \frac{1}{2}$ $= 7 \frac{2}{3}$	M1 M1 A1 B1 M1 M1 A1 M1 A1 B1	For equating Or equivalent For both values For both Accept if c is missing Accept 32.33 or equivalent
		10mks	
24.	$3x + 2.5y \leq 600$ $x \leq 100$ $y \geq 80$ Solution (100, 120) 100 type A and 120 type B garments Profit $= 80 \times 100 + 60 \times 120$ $= \text{sh. } 15,200$	10mks B1 B1 B1 B1 B1 B1 B1 B1 M1 A1	Or equivalent 3x+2.5y ≤ 600 drawn x ≤ 100 drawn y ≥ 80 drawn
		10mks	